Ayush Agrawal

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Summary

DP-600 certified Fabric Data Engineer with hands-on experience designing end-to-end ETL pipelines, lakehouse architectures, and data warehouses using Microsoft Fabric, PySpark, Azure Data Factory, and Power BI. Strong in incremental ingestion, medallion architecture, dimensional modeling, pipeline orchestration, and data quality frameworks for enterprise analytics workloads.

Skills

Programming & Querying: Java, Python, TypeScript, MySQL, SQL
Data Engineering & Platforms: Microsoft Fabric, Azure Data Factory
Big Data & Data Processing: PySpark, Incremental & CDC Data Loads, Delta Tables, Data Validation & Quality Checks
Data Modeling & Warehousing: Dimensional Modeling, Star Schema, Snowflake Schema, Medallion Architecture
Cloud & Dev Tools: Microsoft Azure, Git, Maven, VS Code

Experience

Software Engineering Intern **Jan 2026 - July 2026**
IALTruist LLC, Remote *Tools Used: Python, pytest, TypeScript, Playwright*

- Developed an **AI-assisted UI test generation POC** using Playwright, TypeScript, and Codex/MCP that converts plain-language prompts into test specs, with a hash-based registry to regenerate only changed prompts; covered 90% application modules.
- Built a Swagger/OpenAPI-driven **API test generation framework in Python** that auto-creates dependency-ordered CRUD suites, resolves runtime IDs, and reuses authenticated sessions across tenant/workspace contexts.
- Added structured reporting (Allure, HTML, JSON) and AI-based failure analysis to speed up debugging of automated test runs.
- Implemented UI enhancements on the company website using React, TypeScript, and Material UI.

Data Intern **May 2025 – Aug 2025**
Techment Technology, Bhilai *Tools Used: Microsoft Fabric, Azure, Power BI*

- Built **end-to-end ETL pipelines in Microsoft Fabric** to ingest transactional data from on-prem MySQL into OneLake for enterprise reporting.
- Implemented **incremental ingestion and schema validation frameworks,** improving pipeline reliability by 40%.
- Optimized **PySpark notebooks and Dataflow Gen2 pipelines,** reducing Power BI refresh latency by 35%.
- Implemented pipeline orchestration and failure monitoring using Fabric Data Pipelines.

Projects

Website Analytics ETL & Pipeline | *Microsoft Fabric, PySpark, SQL, Power BI*

- Designed a **full-scale ETL pipeline** to ingest raw website logs from multiple MySQL sources into Fabric Lakehouse.
- Implemented Bronze → Silver → Gold transformation layers using PySpark for cleansing, deduplication, and enrichment.
- Built incremental ingestion pipelines with schema drift handling and validation checks. Modeled dimensional star schema for optimized analytical queries. Exposed curated Gold tables to Power BI via a Semantic model for business reporting.
- Implemented orchestrated pipelines with scheduled refresh and failure handling.

Supply Chain Monitoring ETL & Pipeline | *Microsoft Fabric, PySpark, SQL, Power BI*

- Designed a **full supply-chain pipeline** to track inventory, shipments, vendor performance, and stock movement.
- Built automated ingestion pipelines with logging and failure monitoring using Data Pipelines and stored procedures.
- Designed multipage dashboards with slicers and filters, helping stakeholders drill down by region, product category, and shipment timelines for inventory and delivery monitoring.
- Implemented Semantic model and curated Gold layer enabling inventory, shipment delay, and supplier performance analytics.

Achievements & Certifications

- Microsoft Certified: Fabric Analytics Engineer Associate (DP-600).
- Awarded Student Excellence Award by the Department of Computer Science and Engineering
- Dean's List Excellence in Academics Certificate by Faculty of Engineering.
- Awarded Scholarship by the Directorate of Student Welfare.

Education

B.Tech Computer Science & Engineering **Aug 2022 – July 2026**
Manipal University Jaipur, Jaipur *GPA: 8.9*